

Dynamic and static UWB comparison

FlexTDOA, Native DS-TWR and Passive DS-TWR

Field campaign of August 6, 2026

Platform: 5 ESP32-S3 + DW3000 modules, Raspberry Pi collector
Topology: M1 tag; M2, M3, M4 and M5 anchors
Radio: UWB channel 9; configured 40 MHz SPI
Firmware branch: `imu-500hz-fusion`
Evidence: six complete immutable JSONL captures plus replay summaries

Main result. All three protocols produced usable unfiltered dynamic trajectories. FlexTDOA delivered the highest independent dynamic rate and the best static local precision. Native DS-TWR had the shortest worst-case dynamic gap. Passive DS-TWR remained usable, but its overlapping publications must be reduced to independent frames for a fair rate comparison.

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1 Executive summary

Evidence boundary. Stream rates, uptime gaps, sample counts, fusion acceptance and reset counts are direct diagnostics. Absolute UWB-RTK disagreement is only indicative: RTK was not an independent survey, clocks are associated in software, antenna lever arms are unknown, and the UWB frame is registered through RTK anchor geometry.

Passive DS overlapping raw windows remain visible in the telemetry rate, while all comparative precision and trajectory metrics use independent frames. Interleaved records marked `imu_fused=true` are excluded from raw UWB metrics. Outliers are retained.

The dynamic independent rates were **25.31 positions/s** for FlexTDOA, **21.74 positions/s** for Native DS-TWR, and **16.01 positions/s** for Passive DS-TWR. Passive DS emitted 48.73 dashboard events/s, but those include overlapping windows.

The static raw local RMS values were **2.59 cm** (FlexTDOA), **4.87 cm** (Native DS-TWR) and **4.00 cm** (Passive DS-TWR). These are precision around each capture median, not absolute accuracy.

The stationary tag RTK centers differed by 1.583 m between blocks, invalidating an absolute cross-protocol static ranking. IMU transport was healthy at roughly 119–127 telemetry samples/s, but all six replays reported zero yaw-alignment updates. This campaign therefore does not demonstrate acceleration-derived position improvement.

2 Scope and questions

The campaign asks how many independent positions each protocol delivers, where stream gaps occur, how stable the unfiltered stationary position is, and whether the first IMU-assisted replay improves the result without hiding raw UWB.

The July 26 report used known static geometry and could make a direct accuracy statement. Here the anchors and tag are tied to mobile RTK-derived geometry and software time association, so the evidence boundary is deliberately stricter.

3 System and protocol background

3.1 FlexTDOA

The tag is UWB receive-only. Anchor transmissions form directed time-difference observations and the tag solves from a recent coherent observation set. Radio airtime is largely independent of the number of passive tags.

3.2 Native DS-TWR

The active tag exchanges poll, response and final messages with every anchor, obtains four absolute ranges and solves a coherent frame. Airtime grows with active tag count, but frame semantics are direct.

3.3 Passive DS-TWR

The tag listens to anchor-to-anchor DS-TWR exchanges. Rolling windows may produce several publications from shared measurements, so dashboard events and independent frames are intentionally reported separately.

4 Test configuration and captured evidence

The dynamic walks lasted 91–96 s and the stationary blocks approximately 180 s. The selector uses the final walk and the RTK-fixed Raspberry static capture for each protocol.

Protocol	Mode	Capture ID	duration s	events	independent	indep Hz	coverage pct	complete
FlexTDOA	dynamic	walk_flex_final	91.4	2312	2312	25.31	98.3	yes
Native DS-TWR	dynamic	walk_native_final	91.1	1981	1981	21.74	99.4	yes
Passive DS-TWR	dynamic	walk_passive_final	95.7	4664	1532	16.01	98.7	yes
FlexTDOA	static	rpi_static_flextdoa_rtkfixed_180s	180.1	5138	5138	28.54	100.0	yes
Native DS-TWR	static	rpi_static_native_rtkfixed_180s	180.0	4040	4040	22.44	100.0	yes
Passive DS-TWR	static	rpi_static_passive_rtkfixed_180s	180.1	12557	4185	23.24	99.9	yes

Every selected block has a terminal `capture_end`. All raw data are archived losslessly and identified by SHA-256.

5 Methodology

5.1 Population and continuity

No error-based trimming or temporal smoothing is applied. Rates use full capture wall time. Gaps are positive differences between unique ESP32 uptime values; P50/P95/P99, maximum and counts above 100 ms are retained together.

5.2 Static precision

Radial displacement is computed around each block's component-wise position median. CEP50, RMS, P95, P99, maximum, 2DRMS and first-to-last-decile drift use all independent samples. They describe repeatability, not offset from truth.

5.3 RTK audit

Only quality-4 fixes enter the audit. WGS84 is converted through ECEF to a common ENU frame. A proper 2-D rotation and translation is fitted from at least three anchor centers; scale is fixed. Tag matches must be within 100 ms. Anchor-fit residual and association timing accompany every tag error.

6 Dynamic results

6.1 Raw trajectories

The walks are separate physical trials, not synchronized repeats. The panels therefore show continuity and gross geometry rather than point-by-point route agreement.

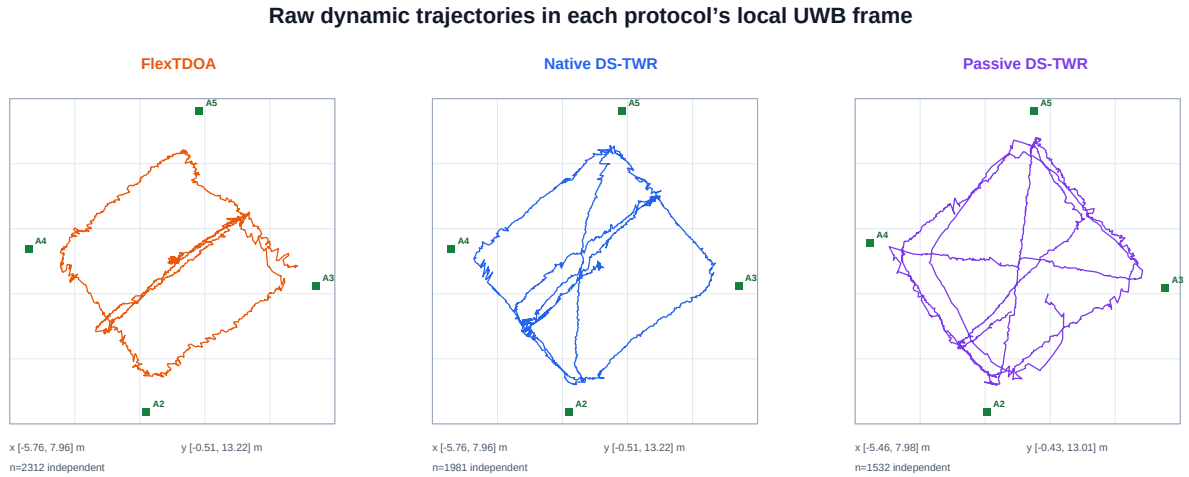


Figure 1: Raw independent dynamic trajectories in each local UWB frame. Green markers are anchors. Different panels represent different walks.

6.2 Update rate and gaps

Protocol	duration s	events	independent	event Hz	indep Hz	gap P95 ms	gap max ms	step P95 m/s
FlexTDOA	91.4	2312	2312	25.31	25.31	80	520	4.44
Native DS-TWR	91.1	1981	1981	21.74	21.74	60	200	5.11
Passive DS-TWR	95.7	4664	1532	48.73	16.01	70	1610	4.19

Native DS-TWR had the best dynamic worst-case continuity (200 ms maximum). FlexTDOA produced the highest independent rate but one 520 ms gap. Passive DS had a useful typical cadence yet one 1.61 s interruption and the most gaps over 100 ms. Its recovery is operationally useful, but rare long gaps remain the first diagnostic target. Step speed is a sample-to-sample trajectory diagnostic, not measured walking speed.

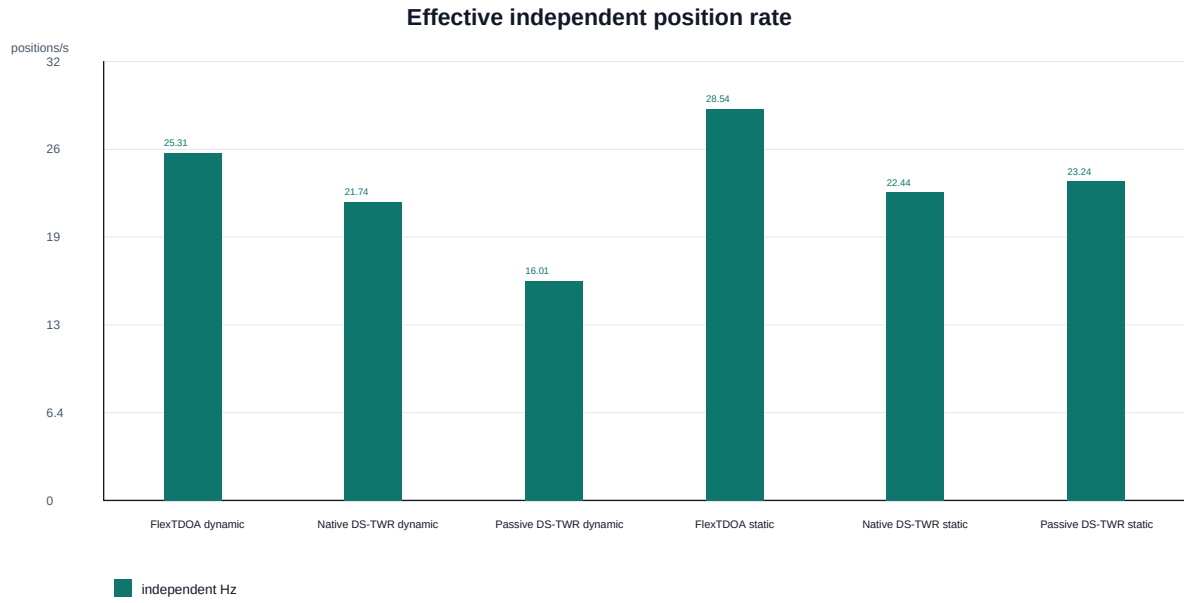


Figure 2: Independent position rate in the selected dynamic and static blocks. Passive DS all-event publications are not substituted for independent frames.

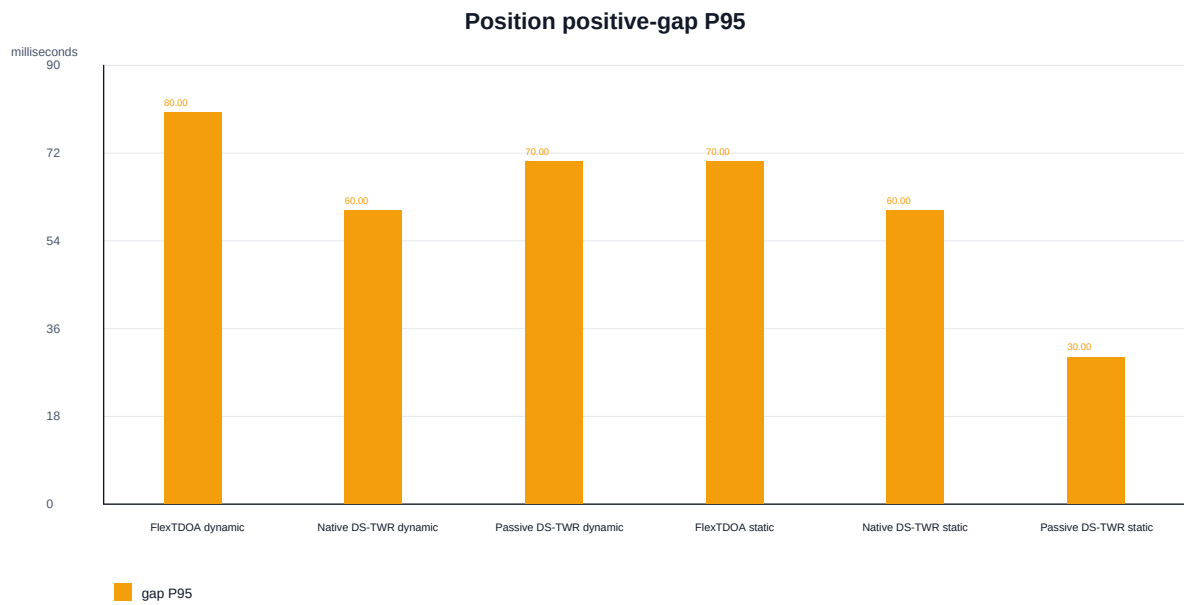


Figure 3: P95 positive gap between unique position uptimes. Maximum gaps and counts remain in the tables.

6.3 Complete continuity audit

Protocol	Mode	event Hz	indep Hz	P50 ms	P95	P99	max	>100 ms
FlexTDOA	dynamic	25.31	25.31	30	80	172	520	54
Native DS-TWR	dynamic	21.74	21.74	40	60	90	200	6
Passive DS-TWR	dynamic	48.73	16.01	10	70	180	1610	151
FlexTDOA	static	28.54	28.54	30	70	90	170	21
Native DS-TWR	static	22.44	22.44	40	60	70	100	0
Passive DS-TWR	static	69.74	23.24	10	30	130	460	183

7 Static results

7.1 Raw local precision

Protocol	N	CEP50 cm	RMS cm	P95 cm	P99 cm	max cm	2DRMS cm	drift cm
FlexTDOA	5138	2.02	2.59	4.30	5.27	55.12	5.17	0.10
Native DS-TWR	4040	1.89	4.87	14.51	21.17	29.56	9.74	0.36
Passive DS-TWR	4185	2.72	4.00	8.19	10.74	14.40	7.89	6.55

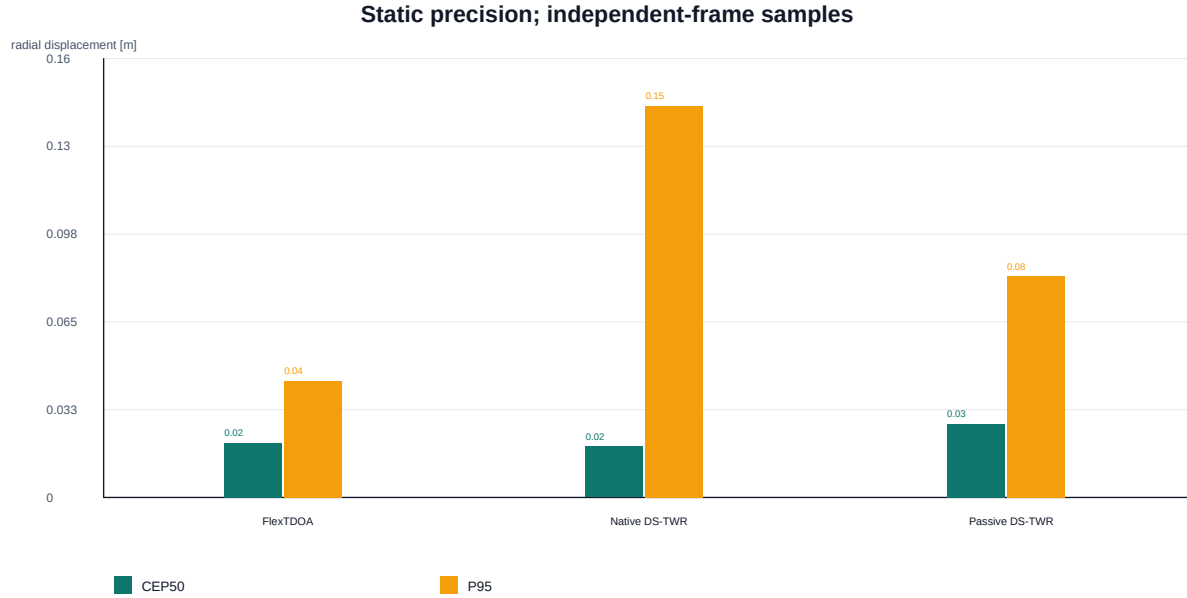


Figure 4: Static precision around each block’s own median. All independent raw samples are included.

FlexTDOA has the smallest static RMS and P95. Its isolated 55.12 cm maximum is retained rather than trimmed. Native DS has the broadest P95, while Passive DS lies between them on RMS and has the smallest maximum. One stationary point cannot characterize GDOP or spatial bias across the anchor polygon.

The RTK Fixed clouds are individually tight: their within-block P95 horizontal spreads are approximately 0.85 cm, 1.15 cm and 2.02 cm for FlexTDOA, Native DS and Passive DS respectively. The common-ENU view nevertheless separates the Passive DS block by about 1.58 m. High short-term RTK precision therefore does not imply cross-session correctness.

7.2 RTK cross-capture consistency gate

The stationary RTK center’s maximum pairwise separation is **1.583 m**, versus a 0.250 m gate. **static absolute UWB-RTK errors are audit-only and invalid for cross-protocol ranking.** FlexTDOA and Native DS centers differ by about 6.5 mm, whereas the Passive DS block differs by approximately 1.58 m.

8 IMU capture and fusion assessment

The sensor request was configured at 200 Hz; the collector observed roughly 119–127 events/s. This is a telemetry result and does not prove that every hardware sample reached the Raspberry Pi.

Static raw position clouds about each capture median

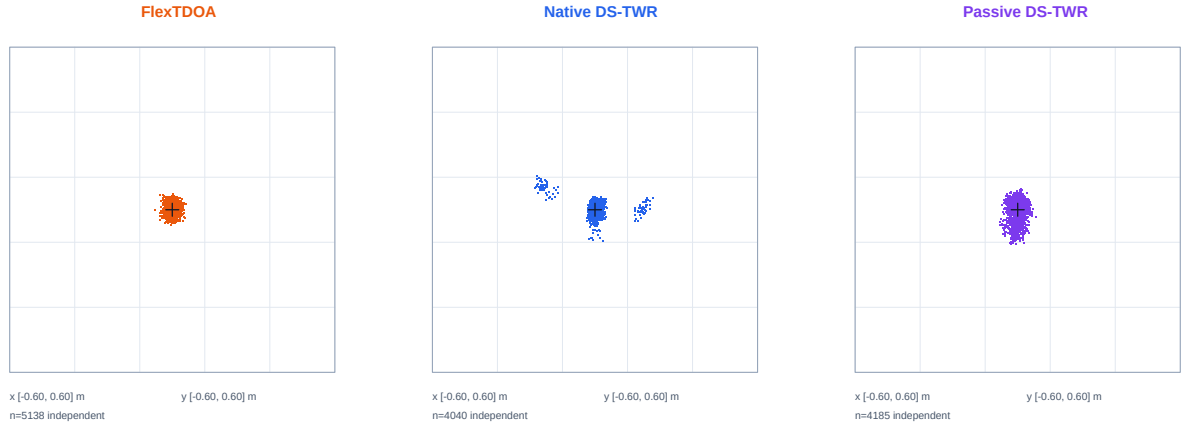


Figure 5: Static raw position clouds centered on their own medians and drawn at one common scale. The cross is the median.

GPS RTK Fixed tag clouds and cross-capture centers

Top: each block centered on its own median. Bottom: the same fixes in one common ENU frame.



Figure 6: GPS RTK Fixed tag solutions. Top: each protocol block is centered on its own median with one shared centimetre scale, showing within-block precision. Bottom: the same fixes and medians in the common ENU frame with equal axis scale, exposing the 1.58 m cross-capture reference jump.

Protocol	Mode	IMU events	observed Hz	valid pct	gap P95 ms	gap max ms
FlexTDOA	dynamic	10869	119.0	100.0	20	50
Native DS-TWR	dynamic	11405	125.2	100.0	10	50
Passive DS-TWR	dynamic	11410	119.2	100.0	30	80
FlexTDOA	static	22871	127.0	100.0	20	50
Native DS-TWR	static	22865	127.0	100.0	20	40
Passive DS-TWR	static	22865	127.0	100.0	20	80

8.1 Replay result

Replay acceptance, outliers and resets are direct diagnostics. RTK columns are secondary evidence because repeated output matches do not create independent fixes.

Protocol	Mode	accept pct	outliers	resets	raw RTK m	fused RTK m	delta mm	yaw updates	anchor fit m
FlexTDOA	dynamic	98.6	7	2	1.095	1.097	2.0	0	0.764
Native DS-TWR	dynamic	98.7	26	1	2.168	2.176	8.3	0	0.400
Passive DS-TWR	dynamic	88.0	90	27	3.436	3.437	0.9	0	0.003
FlexTDOA	static	98.9	0	0	0.111	0.109	-2.1	0	0.073
Native DS-TWR	static	98.6	57	0	0.535	0.534	-0.8	0	0.684
Passive DS-TWR	static	87.6	0	0	1.490	1.490	0.0	0	0.004

All replays have zero yaw-alignment updates, so body-frame horizontal acceleration cannot safely be rotated into the map. Dynamic raw/fused RMSE changes only by millimetres. The fused stream is therefore continuity-oriented prediction/correction, not demonstrated inertial dead reckoning. Raw UWB stays separate and is the basis of this protocol comparison.

9 UWB–RTK association audit

The following values are retained for transparency, not promoted to surveyed accuracy. Debiased RMSE removes median tag residual and is a shape diagnostic.

Protocol	Mode	pairs	time P95 ms	RMSE m	P95 m	debiased m	anchor fit m	alignment
FlexTDOA	dynamic	23	36.3	1.158	2.135	0.989	0.763	poor_alignment_do_not_claim_accuracy
Native DS-TWR	dynamic	27	27.2	2.164	3.433	1.263	0.390	poor_alignment_do_not_claim_accuracy
Passive DS-TWR	dynamic	19	63.6	3.298	4.756	1.903	0.268	usable_with_limitations
FlexTDOA	static	119	31.6	0.087	0.115	0.026	0.006	usable_with_limitations
Native DS-TWR	static	119	28.9	0.092	0.122	0.055	0.110	poor_alignment_do_not_claim_accuracy
Passive DS-TWR	static	101	63.3	1.471	1.539	0.048	0.091	usable_with_limitations

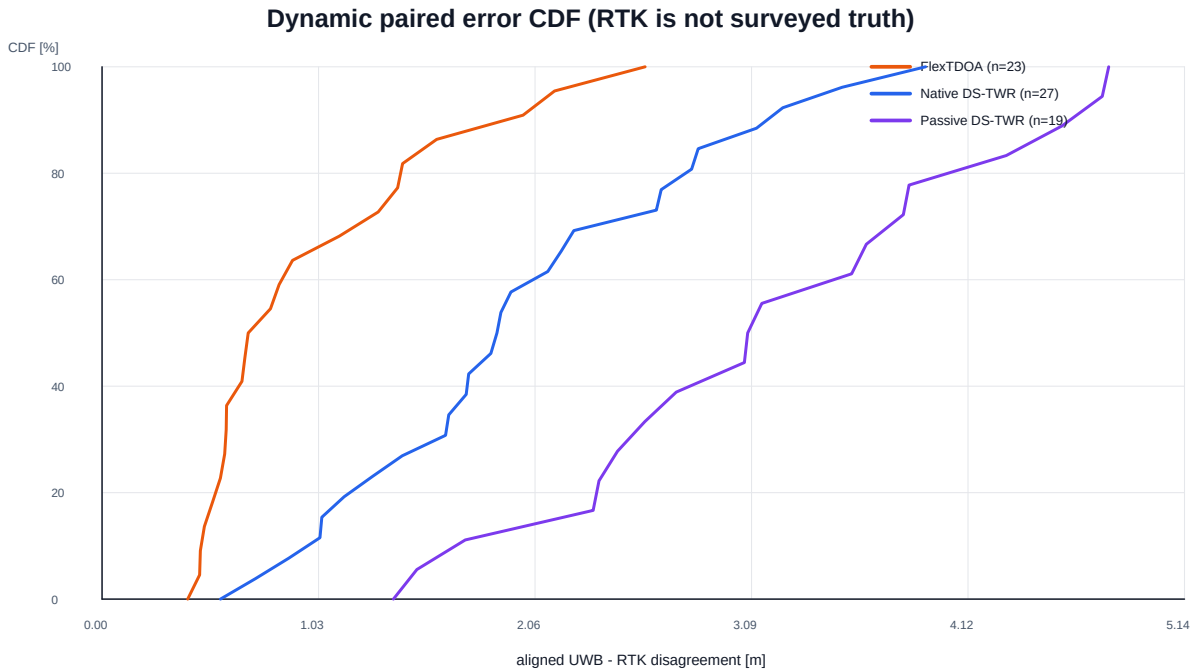


Figure 7: Dynamic UWB–RTK disagreement after anchor-based rigid registration. This is not surveyed position error.

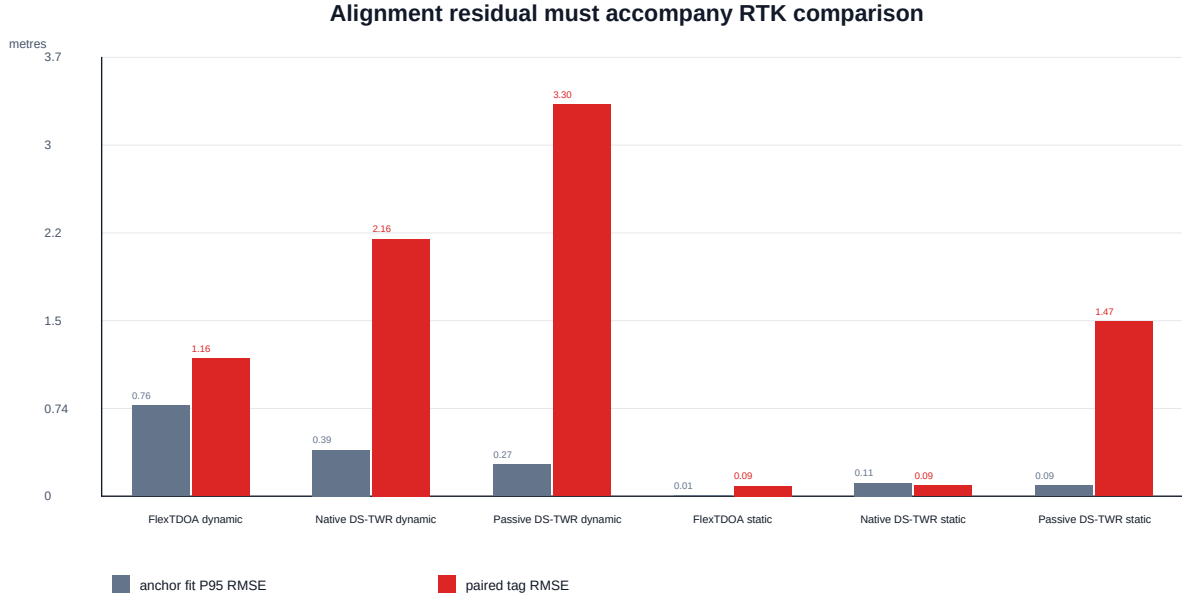


Figure 8: Anchor-registration residual beside paired tag disagreement. Registration uncertainty limits tag interpretation.

Dynamic paths and anchor fits differ by capture. Passive DS dynamic geometry is partly GNSS-derived, making its registration non-independent. The RTK audit is still useful for gross frame errors and timing failures.

10 Protocol interpretation and selection

10.1 FlexTDOA

Best independent dynamic rate and static raw precision in this campaign, with a receive-only scalable tag. The 520 ms dynamic maximum gap is the principal caveat. The walks do not justify another protocol rewrite.

10.2 Native DS-TWR

Clearest coherent-frame semantics and best dynamic maximum gap. Static P95 is wider here, so per-link bias/antenna-delay validation remains worthwhile. It is the preferred deterministic baseline for few active tags.

10.3 Passive DS-TWR

Successfully follows motion without tag transmissions. Only one third of its rolling publications are independent in these captures; the 1.61 s dynamic gap and higher replay reset count keep recovery diagnostics as the priority.

Criterion	FlexTDOA	Native DS-TWR	Passive DS-TWR
Tag radio role	receive-only	active exchanges	receive-only
Dynamic independent rate	highest	middle	lowest after overlap removal
Dynamic worst gap	520 ms	best: 200 ms	1.61 s
Static raw precision	best RMS/P95	broadest P95	middle
Scaling with tag count	strongest	airtime grows	strong
Recommendation	primary scalable mode	coherent baseline	continue recovery tuning

11 Limitations

- RTK is an in-system comparison reference, not independently surveyed truth.
- Measurement time subtracts receiver-reported fix age from collector time; clocks are not hardware synchronized.
- GNSS and UWB antenna phase centers have unknown lever arms.

- Dynamic routes are not repeated surveyed paths; static/dynamic blocks are single trials.
- One static point does not cover edge geometry, NLOS, orientation or spatial bias.
- Collector rate is end-to-end telemetry, not direct DW3000 PHY throughput.

12 Recommended next tests

1. Repeat every protocol on one marked route with identical orientation and synchronized markers.
2. Interleave static blocks at the center, edges and outside the polygon.
3. Keep one continuous RTK session and log correction age/reset state across protocol changes.
4. Add hardware-correlated timestamps for latency claims.
5. Validate body-to-map yaw before enabling horizontal acceleration corrections.
6. For Passive DS, capture focused diagnostics around every gap above 250 ms.

13 Preliminary findings

- FlexTDOA is the strongest overall scalable result in this campaign.
- Native DS-TWR remains the clean coherent-frame and continuity baseline.
- Passive DS-TWR is usable, but overlap-aware rate accounting and rare-gap recovery remain essential.
- The RTK consistency failure prevents a static absolute-accuracy ranking; local precision alone is ranked.
- IMU transport is healthy, but acceleration-assisted accuracy is not yet demonstrated.

14 Reproduction and audit files

The directory contains `analysis_summary.json`, capture/replay/alignment CSV files, SVG/PDF plots, six loss-less `data/*.jsonl.xz` captures, `raw_data_manifest.csv`, SHA256SUMS and this TeX source.

```
python tools/ubw_dynamic_static_report.py --require-rpi-static
```

No raw capture is modified during report generation.